

Nuclear radiation-proof robot

Toshiba has come up with a tetrapod robot that can withstand high levels of radiation at damaged nuclear plants. It can carry out both investigative and recovery work.

According to the company, the new robot integrates a camera and a dosimeter and can investigate the condition of nuclear power plants by remote-controlled operation. The multiple joints of its legs are controlled by a dedicated movement algorithm that enables the robot to walk on uneven surfaces, avoid obstacles and climb stairs, securing access into areas that are challenging to reach by wheeled robots or crawlers.

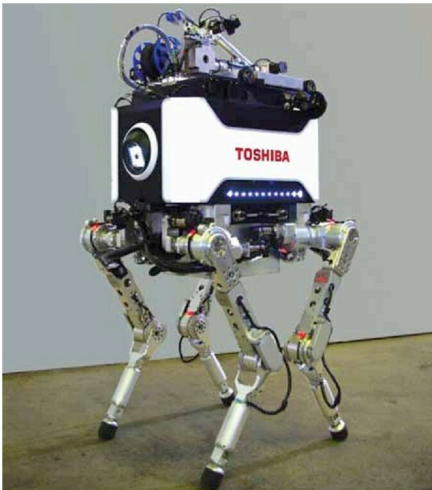
"The robot also has a folding arm that can release a companion smaller robot that mounts a second camera. This can be launched from the main robot and positioned to take images of narrow places and any equipment behind them, and tubes and other places that are too small for the robot to enter. It is connected to the main robot by a cable"—the company explained in a release.

Robot that walks, swims and flies

Engineers at the Illinois Institute of Technology have designed a robot that can walk on earth, swim in the water and fly in the sky. HyTAQ, as it is fondly known as, is a Hybrid Terrestrial and Aerial Quadrotor robot.

The robot is designed by Arash Kalantari and Mathew Spenko. HyTAQ is capable of doing terrestrial and aerial locomotion equally well. When it walks, it needs to budge the rolling resistance. However, it consumes less power in this motion, which increases its operating time.

The robot is capable of moving four times greater distances. The best part is its obstacle avoidance capability; as soon as it finds an obstacle in its way, it just flies to overcome it.



Toshiba's quadruped robot for nuclear sites inspection

Once the robot enters flight mode, its configuration changes in such a way that its four actuators provide the desired thrust. The robot can use the same actuator set and control system for terrestrial locomotion as well just by adding a rolling cage to the quadrotor.

Stretchable electronics will soon be reality

In the future, we are going to have stretchable biomedical, wearable, portable and sensory devices. Bingqing Wei from University of Delaware is helping companies like IBM, Sony and Nokia to make this a reality in their products.

"Advances in soft and stretchable substrates and elastomeric materials have given rise to an entirely new field," Wei said. Such stretchable devices will also need stretchable power source. Wei added, "Rechargeable and stretchable energy storage devices, also known as supercapacitors, are urgently needed to complement advances currently being made in flexible electronics."

The research team at University of Delaware is progressing in developing scalable, stretchable power sources to aid the kind of application with the use of carbon nanotube macrofilms, polyurethane membranes and organic electrolytes.